Fraud Detection System for Digital Payments and Banking Transactions

Course: Artificial Intelligence

Assignment: PEAS Framework and Task Environment Analysis

Reference: Russell & Norvig – Artificial Intelligence: A Modern Approach, Chapter 2

Student Name: Memoona Lashari
Roll No: 2K23/CSE/82
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Introduction

For this assignment I picked the Fraud Detection System used by banks and digital payment providers to identify suspicious financial transactions before they are completed. Every time someone uses a debit card, credit card, online banking service, or mobile payment application, the system analyzes the transaction and decides whether it appears legitimate or fraudulent.

I selected this system because it is one of the most common real-world examples of an intelligent agent operating under uncertainty. The system has to make decisions within milliseconds while processing huge numbers of transactions every day. At the same time, fraudsters continuously change their methods, making the environment highly challenging and constantly evolving.

The system observes information such as transaction amount, location, customer behavior, device information, and historical activity. Based on this information it decides whether to approve the transaction, request additional verification, or block the payment entirely. This makes it a strong example of an intelligent agent with clear goals, inputs, outputs, and performance measures.

Task 1: PEAS Framework

The system I am analyzing is a Fraud Detection System deployed within banking networks, digital payment platforms, and financial institutions to identify and prevent fraudulent transactions in real time.

Performance Measures

These are the things the system tries to do well.

1. Fraud Detection Rate

This measures the percentage of fraudulent transactions successfully identified and blocked by the system. A high detection rate means better protection against financial losses.

2. False Positive Rate

This measures the percentage of legitimate transactions incorrectly flagged as fraudulent. Keeping this value low is important for maintaining customer satisfaction.

3. Transaction Review Latency

This measures the average time required to analyze and classify a transaction. Fast decisions are necessary for seamless payment processing.

4. Financial Loss Prevented

This measures the total amount of money saved by stopping fraudulent activities before they are completed.

Environment

The Fraud Detection System operates inside global digital payment networks including online shopping websites, banking applications, ATMs, card processing systems, and mobile payment platforms. These systems run continuously twenty-four hours a day and process millions of transactions from customers around the world.

The environment includes customers, merchants, banks, payment processors, regulators, and fraudsters. Transaction patterns change throughout the year because of holidays, shopping events, and economic conditions. In addition, security regulations and compliance requirements vary across different countries.

Because of these factors, the system must operate in a complex and constantly changing environment.

Actuators

These are the ways the Fraud Detection System performs actions and affects its environment.

• Transaction Decision Engine

Approves, reviews, or declines transactions depending on the calculated risk level.

• Step-Up Authentication Trigger

Requests OTP verification, biometric authentication, or additional security checks when suspicious activity is detected.

• Fraud Investigation Alert

Creates investigation tickets and alerts for human fraud analysts.

• Account Control Module

Temporarily freezes accounts or limits transaction capabilities when severe fraud risk is identified.

• Regulatory Reporting Module

Automatically generates reports required by financial regulators and compliance authorities.

Sensors

These are the inputs used by the system to understand and evaluate transactions.

• Transaction Metadata

Collects transaction amount, merchant information, location, transaction type, and timestamp.

• Device Fingerprinting

Identifies whether the transaction originates from a recognized customer device.

• Behavioral Biometrics

Monitors typing speed, swipe patterns, mouse movements, and other user interaction behaviors.

• IP Reputation Detector

Evaluates network addresses and geographic locations for suspicious activity.

• Transaction Velocity Counter

Detects unusually high transaction frequencies occurring within short periods of time.

• Transaction History Database

Analyzes customer behavior and spending patterns from previous transactions.

• External Threat Intelligence Feed

Receives real-time updates about emerging fraud schemes and malicious entities.

Task 2: Environment Classification

Dimension	Classification	Justification
Fully Observable vs. Partially Observable	Partially Observable	The system can observe transaction information but cannot directly determine customer intent or whether credentials have been stolen.
Single-Agent vs. Multi-Agent	Multi-Agent	Customers, banks, payment processors, and fraudsters all interact simultaneously within the environment.
Deterministic vs. Stochastic	Stochastic	Similar transactions may produce different outcomes because of hidden factors unavailable to the system.
Episodic vs. Sequential	Sequential	Each fraud decision updates customer risk profiles and influences future transaction assessments.
Static vs. Dynamic	Dynamic	Fraud techniques continuously evolve, requiring the system to adapt to changing attack strategies.
Discrete vs. Continuous	Continuous	Risk scores, transaction amounts, and behavioral indicators exist on continuous numerical scales.

Task 3: Critical Analysis

3.1 Most Technically Challenging Dimension

In my opinion the hardest part of building this system is dealing with the Dynamic Environment. Fraudsters constantly adapt their strategies whenever security systems become effective at detecting a particular fraud technique.

The challenge is that the fraud detection model cannot simply be trained once and then left unchanged. Engineers must continuously monitor new fraud trends, retrain machine learning models, test updates, and deploy improvements without disrupting normal banking operations.

Another difficulty is adversarial behavior. Fraudsters intentionally attempt to deceive the system and may even try to manipulate training data so that future detection becomes less effective.

Because of these factors, maintaining performance in a dynamic environment requires continuous effort and remains the most technically demanding aspect of fraud detection.

3.2 Utility Function and Trade-Off Analysis

For a Fraud Detection System the main trade-off exists between catching fraudulent transactions and avoiding unnecessary disruption to legitimate customers.

The proposed utility function is:

$$U = \alpha \times \text{Detection\_Rate} - \beta \times \text{False\_Positive\_Rate} - \gamma \times \text{Latency\_Penalty}$$

Where Detection Rate represents successful fraud prevention, False Positive Rate represents incorrectly blocked legitimate transactions, and Latency Penalty represents delays in decision making.

If the weight of Detection Rate is increased significantly, the system becomes much more aggressive. It catches additional fraudulent transactions but also blocks more legitimate customer payments. Customers may become frustrated and lose trust in the institution.

This demonstrates why fraud detection systems must carefully balance security, customer satisfaction, and processing speed. Optimizing one factor too aggressively can negatively affect the others.

Task 4: Structured JSON Representation


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{
  "system_name": "Fraud Detection System for Digital Payments and Banking Transactions",

  "peas": {

    "performance_measures": [
      "Fraud Detection Rate: percentage of fraudulent transactions successfully identified and blocked by the system",
      "False Positive Rate: percentage of legitimate transactions incorrectly flagged as fraud",
      "Transaction Review Latency: average time required to analyze and classify a transaction",
      "Financial Loss Prevented: total monetary value saved by stopping fraudulent activities"
    ],
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    "environment": "Global digital payment networks including online banking,
mobile payments, credit card processing, ATMs, and e-commerce platforms
operating continuously across multiple countries.",

    "actuators": [
      "Transaction Decision Engine: approves, reviews, or declines
transactions based on calculated risk",
      "Step-Up Authentication Trigger: requests OTP, biometric verification,
or additional identity checks",
      "Fraud Investigation Alert: creates cases and notifications for human
fraud analysts",
      "Account Control Module: freezes accounts or limits transaction
capabilities when severe risk is detected",
      "Regulatory Reporting Module: automatically generates compliance
reports required by financial authorities"
    ],

    "sensors": [
      "Transaction Metadata: collects transaction amount, merchant
information, location, and timestamp",
      "Device Fingerprinting: identifies whether the transaction originates
from a recognized customer device",
      "Behavioral Biometrics: monitors typing speed, swipe patterns, and user
interaction behavior",
      "IP Reputation Detector: evaluates network addresses and geographic
locations for suspicious activity",
      "Transaction Velocity Counter: detects unusually high transaction
frequencies within short periods",
      "Transaction History Database: analyzes customer spending behavior from
previous transactions",
      "External Threat Intelligence Feed: receives real-time updates about
emerging fraud patterns and malicious entities"
    ]
  },

  "environment_classification": {

    "fully_observable_vs_partially_observable": {
      "choice": "Partially Observable",
      "justification": "The system can observe transaction information but
cannot directly determine customer intent or whether credentials have been
stolen."
    },

    "single_agent_vs_multi_agent": {
      "choice": "Multi-Agent",
      "justification": "Customers, banks, payment processors, and fraudsters
all interact simultaneously within the environment."
    },

    "deterministic_vs_stochastic": {
      "choice": "Stochastic",
      "justification": "Two transactions with similar characteristics may
produce different outcomes because of hidden factors unavailable to the
system."
    },
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    "episodic_vs_sequential": {
      "choice": "Sequential",
      "justification": "Each fraud decision updates customer risk profiles
and influences future transaction assessments."
    },

    "static_vs_dynamic": {
      "choice": "Dynamic",
      "justification": "Fraud techniques continuously evolve, requiring the
system to adapt to changing attack strategies."
    },

    "discrete_vs_continuous": {
      "choice": "Continuous",
      "justification": "Risk scores, transaction amounts, and behavioral
indicators exist on continuous numerical scales."
    }
  },

  "utility_function": "U = alpha * Detection_Rate - beta *
False_Positive_Rate - gamma * Latency_Penalty"
}
```

References

Russell, S., & Norvig, P. (2022). *Artificial Intelligence: A Modern Approach* (4th ed.). Pearson.

Bolton, R. J., & Hand, D. J. (2002). *Statistical Fraud Detection: A Review*. Statistical Science, 17(3), 235–255.

Phua, C., Lee, V., Smith, K., & Gayler, R. (2010). *A Comprehensive Survey of Data Mining-Based Fraud Detection Research*. Artificial Intelligence Review, 34(1), 1–14.